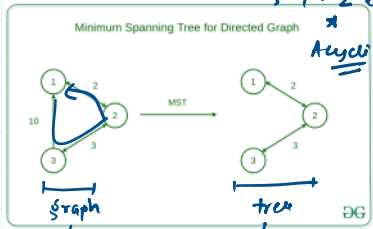


Spanning trees



3 → vertices
3-1 = 2 edges
connected subgraph 's' of graph G(V,E)
is said to be spanning iff

Acyclic

*** sub

- 1) if 's' should contain all vertices of G.
- 2) if 's' should have $|V|-1$ edges, No cycles formed.

* DP

* Basic, Links, Questions

* projects, Assignment *** DSA



document, provide links in order

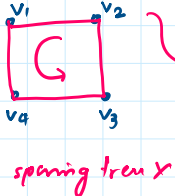
"AI"

↓

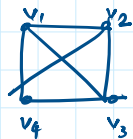
chat

* google map,

* linked list T attend

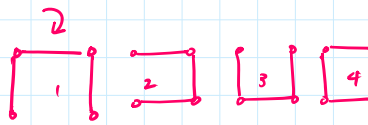


spanning tree
(E = |V|-1)

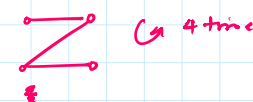


get spanning tree;

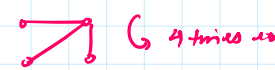
4 combination 1)



4 combination 2)



4 combination 3)



4 combination 4)



16 combinations of spanning tree

Generalized formula of finding spanning tree: # no of vertices = $\frac{n(n-1)}{2}$
4 = $\frac{4(4-1)}{2} = \frac{4 \times 3}{2} = 6$ comb

Kruskal Algorithm:

* Heap
* graph

Heap sorting Algorithm (n log n)

Min Heap

Max Heap

[4, 6, 10, 9, 2] not sorting

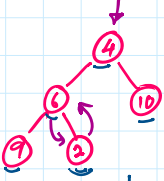
ASC

[2, 4, 6, 9, 10] → sorted

Heap sort + trees.

4, 6, 10, 9, 2

tree (left to right)



min

parent root node should be smaller

max

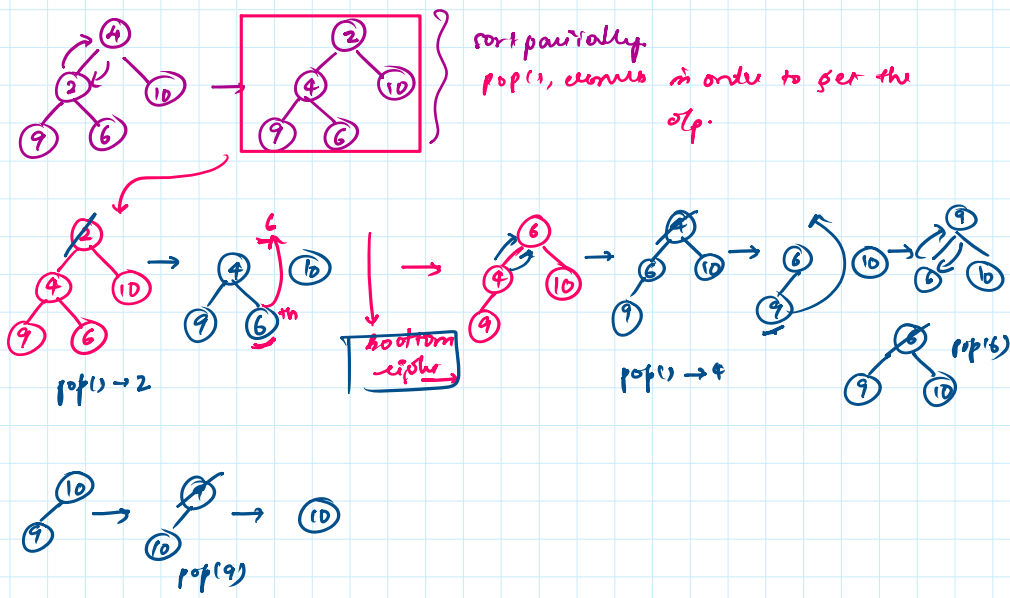
parent root node should be greater

min Heap → $O(n \log n)$ + c.

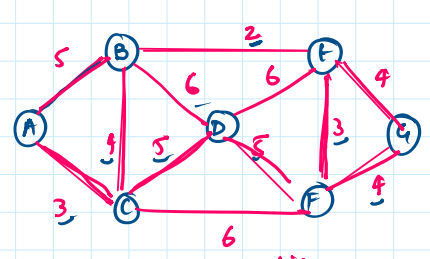
2, 4, 6, 9, 10

7

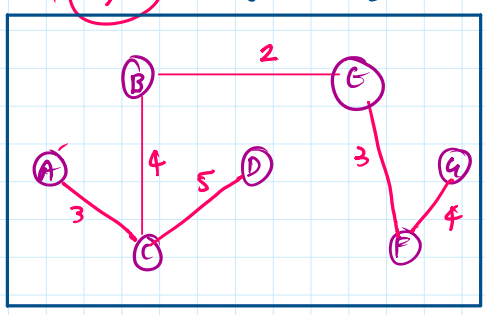
$\rightarrow$ min $\rightarrow$ parent root node should be greater
 8:7 $\rightarrow$ min Heap $\rightarrow O(n \log n)$ + c. $pop = 2, 4, 6, 9, 10$



* Kruskal: 1) min Heap with edges $\rightarrow$ sort (ASC)
 2) take one by one edges $\rightarrow$ graph $\rightarrow$ s-t. Add it in spanning tree
 3) take min span $\rightarrow [V] - 1 = E$, Acyclic.



m.s.t. op.

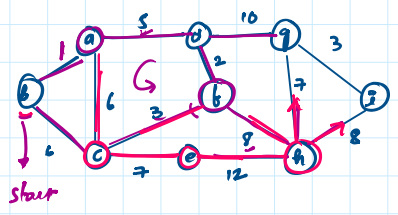


$(E, G, F) \rightarrow$ cycle
 spanning $\rightarrow$ no cycle

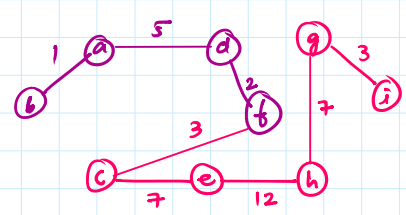
op: (2, 3, 3, 4, 4, 4, 5, 5, 5, 6, 6)
 Asc $\rightarrow$ min heap sorting $\rightarrow O(n \log n)$

$B, C, D, E, F \rightarrow$ cycle
 $B, C, E, F \rightarrow$ cycle

Prim's algorithm



simple Algorithm,
 $(V \rightarrow$ No of vertices.) $E \rightarrow (V-1)$
 * connected Graph.
 kruskal $\rightarrow$ lowest, ASC $\rightarrow$ DESC
 edges, minHeap.



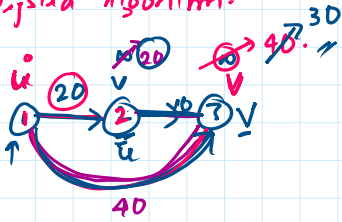
minimum spanning tree from
 prim Algorithm.

Dijkstra Algorithm:

$\rightarrow$ 3D $\rightarrow$ distance or cost



Dijkstra Algorithm:



distance = cost
 $\times$ if $d(u) + c(u,v) < d(v)$
 then: $d(v) = d(u) + c(u,v)$

$$1 \rightarrow 2 \quad d(u) + c(u,v) < d(v)$$

$$0 + 20 < \infty$$

$\boxed{20 < \infty} \rightarrow \text{true: } d(v) = d(u) + c(u,v)$
 $d(v) = 20$

$$1 \rightarrow 3 \quad d(u) + c(u,v) < d(v)$$

$$0 + 40 < \infty$$

$$40 < \infty$$

$$2 \rightarrow 3 \quad d(u) + c(u,v) < d(v)$$

$$20 + 10 < 40$$

$$30 < 40 \rightarrow$$

$$d(v) = d(u) + c(u,v)$$

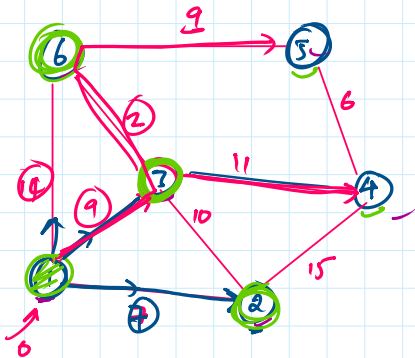
$$d(v) = 0 + 40$$

$$d(v) = 40$$

$$d(v) = d(u) + c(u,v)$$

$$30 = 20 + 10$$

$$\infty > 22$$



- 1, 6, 5 $\rightarrow$ 23
- 1, 2, 4, 5 $\rightarrow$ 28
- 1, 3, 4, 5 $\rightarrow$ 26
- 1, 3, 6, 5 $\rightarrow$ 20

src

destination.

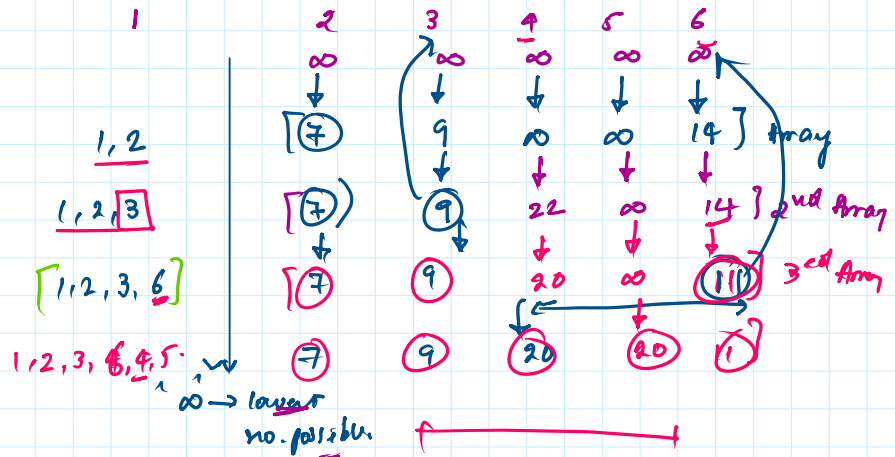


table
 Dijkstra Algo

